

Clinical Protocol Red-Flag Analysis

Case Study: NCT06890338 – Combination Therapy for Advanced Ovarian Cancer

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Study Registry: [ClinicalTrials.gov](https://clinicaltrials.gov)

Trial ID: NCT06890338

Objective

Clinical trial protocols strongly influence the success of a study. Poorly designed protocols can result in recruitment delays, patient dropout, or poor data quality. This analysis evaluates the protocol for NCT06890338 to identify potential operational and data-related risks that could affect trial feasibility, recruitment efficiency, participant retention, and data integrity.

Study Overview

This clinical trial evaluates the anti-tumor activity of a combination therapy consisting of Carboplatin and Mirvetuximab Soravtansine in participants with advanced epithelial ovarian cancer, fallopian tube cancer, or primary peritoneal cancer. Mirvetuximab Soravtansine is a targeted therapy designed to bind to Folate Receptor Alpha, a biomarker commonly expressed in ovarian cancer cells.

The table below summarizes key operational characteristics of the study, including the trial phase, target population, estimated enrollment, and treatment schedule. These factors provide context for evaluating the feasibility and potential operational challenges associated with the protocol.

Study Variable	Description
Study Type	Interventional clinical trial
Phase	Phase II
Estimated Enrollment	140 participants
Population	Adults with advanced ovarian cancer
Treatment Cycle	Every 21 days
Treatment Duration	6–9 cycles depending on response

Protocol Complexity Assessment

Clinical trial complexity influences recruitment and retention. Trials requiring frequent visits, extensive monitoring procedures, or long participation periods can increase patient burden and risk of withdrawal.

The following assessment reviews key factors that contribute to the overall complexity of the protocol.

Risk colour key: High Moderate Low

Complexity Factor	Observation	Risk
Treatment administration	Intravenous infusions required	Moderate
Visit frequency	Multiple monitoring visits	Moderate
Monitoring procedures	Imaging and laboratory tests	High
Study duration	Approximately 3 years	Moderate

Based on these factors, the protocol demonstrates a moderate-to-high level of complexity. Monitoring procedures, treatment administration requirements, and study duration may increase the workload for study sites and the time commitment required from participants.

Protocol complexity score: 7/10. This suggests moderate-to-high operational complexity, which may influence recruitment and retention.



Eligibility Criteria Risk Assessment

Clinical trial eligibility criteria determine which patients can participate in a clinical trial. Although strict criteria can improve scientific precision, they may also reduce the number of patients who qualify for enrollment.

The following assessment evaluates whether the eligibility criteria in this protocol may create recruitment bottlenecks or limit real-world patient representation. The goal is to identify the potential risks that could affect study enrollment and suggest areas where protocol flexibility might improve feasibility.

Criterion	Potential Issue	Impact
FRα expression ≥75%	Requires biomarker testing	Reduced eligible population
ECOG 0–1 only	Excludes weaker patients	Smaller recruitment pool
Stage III–IV only	Limited disease stage eligibility	Slower recruitment

Based on the review of the eligibility requirements, several criteria may narrow the potential participant pool.

The requirement for high Folate Receptor Alpha (FRα) expression requires additional biomarker testing and may exclude patients who otherwise meet clinical criteria. Additionally, restricting participation to ECOG performance status 0–1 removes patients with slightly reduced functional status who may still benefit from treatment.

These criteria may narrow the potential participant pool and could slow recruitment in clinical settings where biomarker testing availability is limited.

Data Collection and Quality Risks

Data quality is critical in clinical trials because regulatory decisions rely heavily on accurate and consistent datasets.

This section evaluates potential sources of data inconsistency within the protocol, particularly those that could lead to data queries, protocol deviations, or delays in database lock.

Data Area	Potential Risk
Imaging assessments	Variability across study sites
Laboratory testing	Differences in biomarker assay procedures
Adverse event reporting	Delayed or incomplete documentation
Manual data entry	Transcription errors

Highlighted rows are the areas the analysis identifies as the most notable data quality risks.

The analysis indicates that imaging assessments and laboratory testing present notable data quality risks. Imaging interpretation may vary between investigators or study sites, potentially affecting outcome measurements such as tumor response assessments. Similarly, differences in laboratory assay procedures could produce inconsistent biomarker results.

To mitigate these risks, centralized imaging review and standardized laboratory procedures are commonly recommended. These approaches help ensure consistency across participating study centers and reduce variability in critical outcome measurements.

Protocol Risk Heatmap

A risk heatmap provides a structured method for evaluating operational and data-related risks in a clinical protocol.

The following matrix summarizes the key risks identified during the protocol review and highlights areas where mitigation strategies may be necessary to ensure smooth trial execution.

ID	Risk Category	Likelihood	Impact	Overall Rating
R1	Strict biomarker eligibility	High	High	High
R2	Frequent monitoring requirements	High	Moderate	Medium-High
R3	Imaging variability	Moderate	Moderate	Medium
R4	Manual data entry errors	Moderate	Moderate	Medium
R5	Patient visit burden	Moderate	High	Medium-High

Overall rating combines likelihood and impact, as shown in the matrix below.

Likelihood × Impact matrix

Likelihood / Impact	Low	Moderate	High
High		R2	R1
Moderate		R3, R4	R5
Low			

R1 Strict biomarker eligibility | R2 Frequent monitoring requirements | R3 Imaging variability | R4 Manual data entry errors | R5 Patient visit burden

The heatmap indicates that strict biomarker eligibility requirements (R1) present the highest combined risk due to both high likelihood and high impact on recruitment. Patient visit burden (R5) and frequent monitoring requirements (R2) also pose moderate-to-high risks that may influence participant retention.

Addressing these risks through operational planning, patient support programs, and improved site training may reduce potential delays and improve trial efficiency.

Sample Data Queries Generated During Data Review

Example Query – Missing Biomarker Result

During routine data review, the Folate Receptor Alpha (FR α) biomarker result for Participant ID 021 was found to be missing in the electronic Case Report Form (eCRF). Since FR α positivity is a required eligibility criterion for this study, confirmation of the biomarker result is necessary to ensure protocol compliance.

Please confirm the Folate Receptor Alpha (FR α) expression result for Participant ID 021 and update the eCRF accordingly or provide clarification if the test was not performed.

Example Query – Out-of-Range Laboratory Value

The ALT value recorded for Participant ID 034 during Cycle 2 appears outside the expected reference range. Please verify the laboratory value.

Risk Mitigation Strategies

Risk mitigation strategies are important to reduce operational challenges and improve study execution. For this protocol, several approaches may help address the identified risks:

- Expanding recruitment sites
- Improving access to biomarker testing
- Standardizing imaging interpretation through centralized review
- Implementing automated data validation checks in electronic data capture (EDC) systems to reduce manual data entry errors and improve data accuracy
- Reducing patient burden through optimized visit schedules

Providing adequate site training and clear study procedures may also help ensure consistent protocol implementation and high-quality data collection.

Conclusion

This protocol analysis identified several operational risks, including strict biomarker eligibility criteria, potential variability in multi-site data collection, and moderate patient burden due to monitoring requirements. Addressing these risks early improves recruitment efficiency, data quality, and overall clinical trial performance.